March 8th, 2026

To: ABC Energy
89 Hay Rowe Street

Subject: Structural Certification for
Installation of Solar Panels

To whom it may concern:

A design check for the subject building was done on the existing roofing and framing systems for the installation of solar panels over the roof. From the site inspections and structural drawings, the existing roof support structures were observed as follows:

Roof A

The roofing material consists of rubber membrane over rigid insulation that is supported by type B, 1.5", corrugated steel deck. Some of the original 2-in gypsum deck were not removed and still exist under the steel deck. The span of the 10B15 steel joists is 20'-0". The 10B15 joists are supported by steel double angles that land on the main bent trusses at 20'-0" o.c. The trusses have an out-to-out width of 80'-0" between the exterior load bearing walls, and a slope of 14 degrees.

Roof B

The roofing material consists of rubber membrane over rigid insulation that is supported by type B, 1.5" corrugated steel deck, observed in two locations and confirmed during roofing replacement. The roof deck is supported by 10B15 steel joists at 5'-0" o.c. to 6'-8" o.c., spanning in the longitudinal direction. The span of the 10B15 steel joists is 20'-0". The 10B15 steel joists are supported by C12x20.7 steel girders on the north border, and 12WF28/14WF30 steel girders between the first and second bays on the north side, with a span length of 20'-0". The 10B15 steel joists on the rest of the roof are supported by steel trusses at 20'-0" o.c., spanning in the transverse direction. The out-to-out width of the trusses is 60'-0". The north 40'-0" of the roof, as well as the south 20'-0", has a close to flat slope, while the middle pitched roof has a slope of 7 degrees.

Roof C

The roofing material consists of rubber membrane over rigid insulation that is supported by type B, 1.5" corrugated steel deck in the north half of the roof, and original 2-in gypsum concrete deck in the south half of the roof. The roof deck is supported by 10B15 steel joists at 6'-8" o.c., spanning in the transverse direction. There are 3 bays of 10B15 steel joists, each with a bay width of 20'-0". The steel joists are supported by 10WF21 steel girders at the exterior, and 12WF28 steel girders in the interior. The span of the steel girders is 20'-0". The roof has a close to flat slope.

According to IEBC 2021, 805.2 Addition or replacement of roofing or replacement of equipment.

"Any existing gravity load-carrying structural element for which an alteration causes an increase in design dead, live or snow load, including snow drift effects, of more than 5 percent shall be replaced or altered as needed to carry the gravity loads required by the International Building Code for new structures. Any existing gravity load-carrying structural element whose gravity load-carrying capacity is decreased as part of the alteration shall be shown to have the capacity to resist the applicable design dead, live and snow loads, including snow drift effects, required by the International Building Code for new structures."

The roof structure then was re-analyzed by applying IBC 2021 code and standards to evaluate the structural integrity under the impact of solar installation.

The existing roof framing systems with the new type B corrugated steel decking are judged to be adequate to withstand the loadings imposed by the installation of the solar panels. No reinforcement is necessary. The existing roof structure can support another 6.0 psf of solar ballast dead load, on top of existing dead and snow loads. The section of roof C with gypsum concrete deck is eliminated from solar installation (see below).

I further certify that all applicable loads required by IBC 2021 with Massachusetts amendments were applied to the solar ballast system and analyzed by the manufacturer. The applicable loads are indicated in the load combination table and the summation of maximum loads indicated on permit plan, which is part of this submittal package. Furthermore, I was informed that the installation crews have been trained to install the solar panels based on the specific roof installation instructions developed by the manufacturer for the solar ballast system. Finally, I accept the certifications indicated by the solar panel manufacturer for the ability of the panels to withstand high wind and snow loads.

Sincerely,

XXXXXXXXXXXXXXXX
Structural Engineer

Load and Load Combination:

Dead Load: Steel Deck

Rubber Membrane + Coverboard = 2.0 psf
Misc (Rigid Insulation, Vapor Barrier, etc) = 3.0 psf
Corrugated Steel Deck = 3.0 psf
Hung Loads = 5.0 psf
Solar Panels = 6.0 psf
Total Dead Load = 2.0 + 3.0 + 3.0 + 5.0 + 6.0 = **19.0 psf**

Dead Load: Gypsum Deck

Rubber Membrane + Coverboard = 2.0 psf
Misc (Rigid Insulation, Vapor Barrier, etc) = 3.0 psf
2" Gypsum Deck = 16.0 psf
Hung Loads = 5.0 psf
Solar Panels = 6.0 psf
Total Dead Load = 2.0 + 3.0 + 16.0 + 5.0 + 6.0 = **32.0 psf**

Snow Load:

Ground Snow Load, P_g:
City of Salem, MA = 50 psf (Mass Building Code, 780 CMR, 10th Ed.)

Flat Roof snow load = $P_f = 0.7\ C_e C_t I\ p_g$ = 0.7 x 1.0 x 1.1 x 1.1 x 50 psf = 42.4 psf (ASCE-7, Eq. 7.3-1)

C_e = exposure factor = 1.0, (ASCE-7, Table 7.3-1), Terrain C, partially exposed
C_t = thermal factor = 1.1 (ASCE-7, Table 7.3-2),
I = importance factor = 1.1 (ASCE-7, Table 1.5-2), occupancy category III

Sloped roof snow load $P_s = C_s\ P_f$ = 1.0 x P_f = 42.4 psf (ASCE-7, Equation 7.4-1)

C_s = slope factor = 1.0 (ASCE-7 Fig. 7.4-1), Cold roofs with C_t = 1.1

⇨ Use **43.0 psf** roof snow load per Mass Building Code 10th Ed.

Wind Speed:

Basic Wind Speed:
City of Salem, MA = 128 mph (Mass Building Code, 780 CMR, 10th Ed.)

Wind Pressure:

Q_z = 0.00256 K_z K_{zt} K_d K_e V^2 (ASCE-7 2016 Eq. 26.10-1)

K_Z = 1.04 (40' building) (ASCE-7 2016 Table 26.10-1)
K_{zt} = 1.0 (ASCE-7 2016 26.8.2)
K_d = 0.85 (ASCE-7 2016 Table 26.6-1)
K_e = 1.0 (ASCE-7 2016 26.9)
V_{utl} = 128 mph (IBC 2018)

Therefore, q_z = 0.00256 x 1.04 x 1.0 x 0.85 x 1.0 x 128^2 = **37.07 psf**

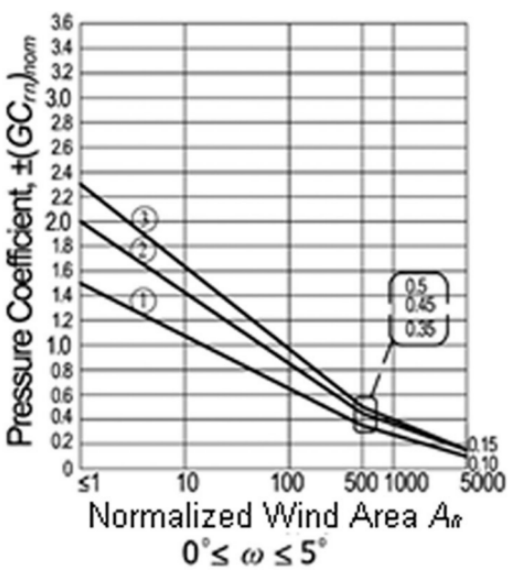
<u>Solar Ballast System Wind Pressure (uplifting): (ASCE-7 2016 Eq. 29.4-5)</u>

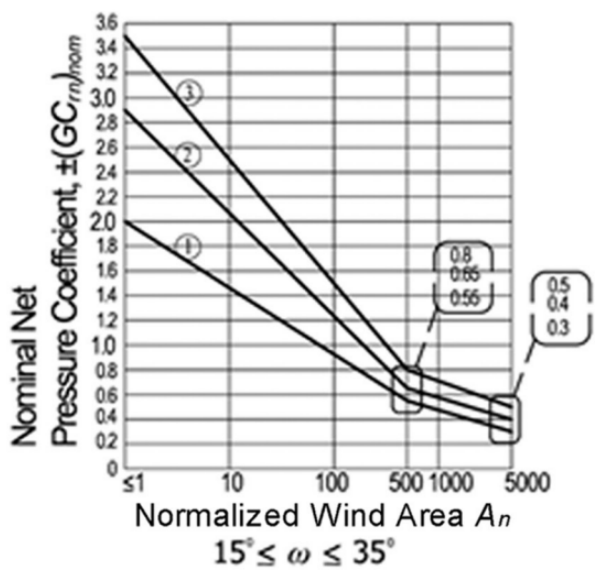
P = q_h (GC_{rn}) = 37.07 psf x -1.546 = -57.31 psf (zone 1)
P = q_h (GC_{rn}) = 37.07 psf x -2.318 = -85.93 psf (zone 2)
P = q_h (GC_{rn}) = 37.07 psf x -2.760 = -102.3 psf (zone 3)

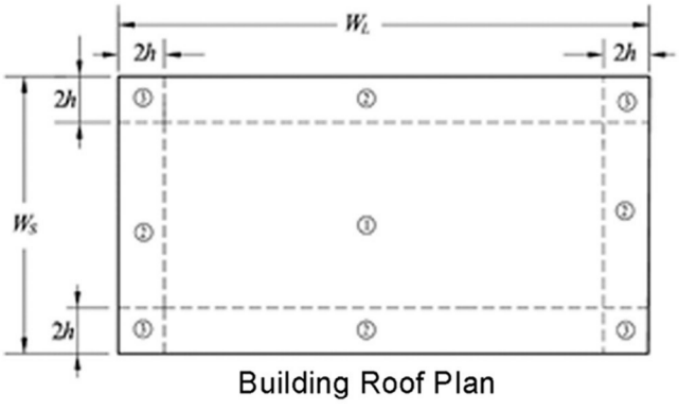
GC_{rn} = (γ_p)(γ_c) (γ_E)(GC_{rn})$_{nom}$ (ASCE-7 2016 Eq. 29.4-6)
= (0.920) (0.8) (1.5) (1.40) = 1.546 (wind zone 1)
= (0.920) (0.8) (1.5) (2.10) = 2.318 (wind zone 2)
= (0.920) (0.8) (1.5) (2.50) = 2.760 (wind zone 3)

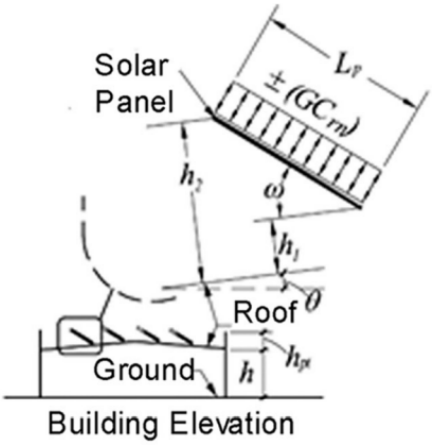
(GC_{rn})$_{nom}$ = 1.40 (zone1), 2.10 (zone 2), 2.50 (zone 3) for normalized area of 9.80 ft^2
γ_P = 0.9 + 1 ft / 50 ft = 0.920 (ASCE-7 2016 29.4.3)
γ_C = 0.8 maximum (ASCE -7 2016 29.4.3)
γ_E = 1.5 for panels within 10' around the edge, or 1.0 for interior (ASCE-7 2016 29.4.4)

a_{pv} = 0.4 (h x W_L)$^{0.5}$ = 0.4 x (50 ft x 380 ft)$^{0.5}$ = 55.14 ft use 50.0 ft
normalized wind area = (1000/a_{pv}^2) x A = (1000/2500) x (24.5 ft^2) = 9.80 ft^2









Seismic Load

S_S = 0.307 (Mass Building Code 10th ed. Table 1604.11)
S_1 = 0.070 (Mass Building Code 10th ed. Table 1604.11)

Spectral Response Acceleration coefficients:

$S_{MS} = F_a S_S = 1.6 \times 0.307 = 0.491$ (IBC Equation 16-20)
$S_{M1} = F_v S_1 = 2.4 \times 0.070 = 0.168$ (IBC Equation 16-21)

Design Spectral Response Acceleration Parameters:

$S_{DS} = 2/3 \times S_{MS} = 2/3 \times 0.491 = 0.327$ (IBC Equation 16-22)
$S_{D1} = 2/3 \times S_{M1} = 2/3 \times 0.168 = 0.112$ (IBC Equation 16-23)

According to IBC Table 1613.3.5 (1) and (2), the seismic design category is "B" and the calculation can be finished by using ASCE-7 equivalent force procedure.

The importance factor, I, for this building is 1.25.

The response modification coefficient, R, is 3.00 according to ASCE-7.

T_L = 6 sec (ASCE Fig 22-14)

$T = C_t h_n^x = 0.028 \times 40^{0.8} = 0.536$ sec (ASCE-7 Equation 12.8-7)

The seismic response coefficient, C_s, is

$C_s = S_{DS} I/R = 0.327 \times 1.25/3.00 = 0.136$